Little Lab Coats

🥔 Potato Osmosis Lab 🥔

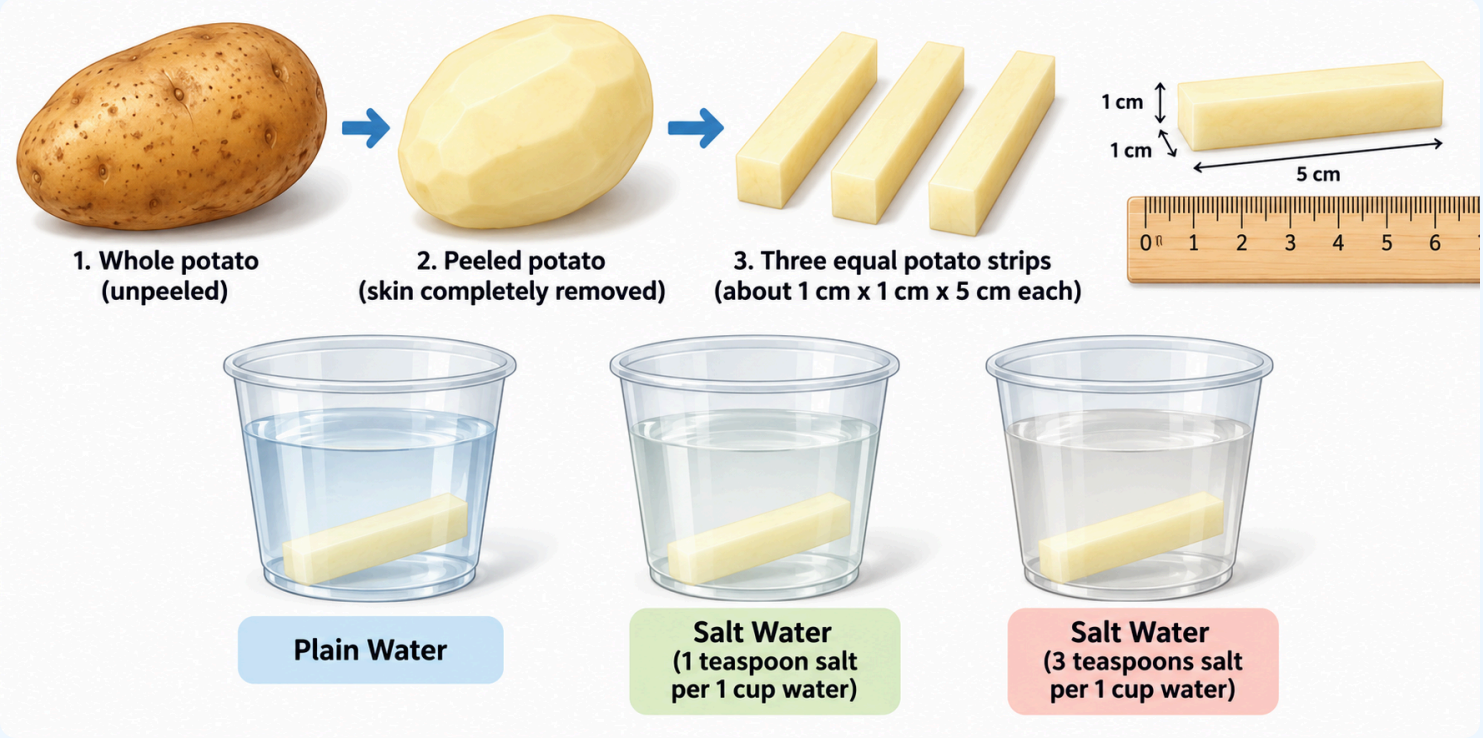
Grade 6 Life Science — Cell Membranes and Water Movement

📚 **Grade:** 6th Grade ⏱️ **Duration:** 30 minutes + 24-hour observation 🔬 **Subject:** Life Science — Cells and Systems

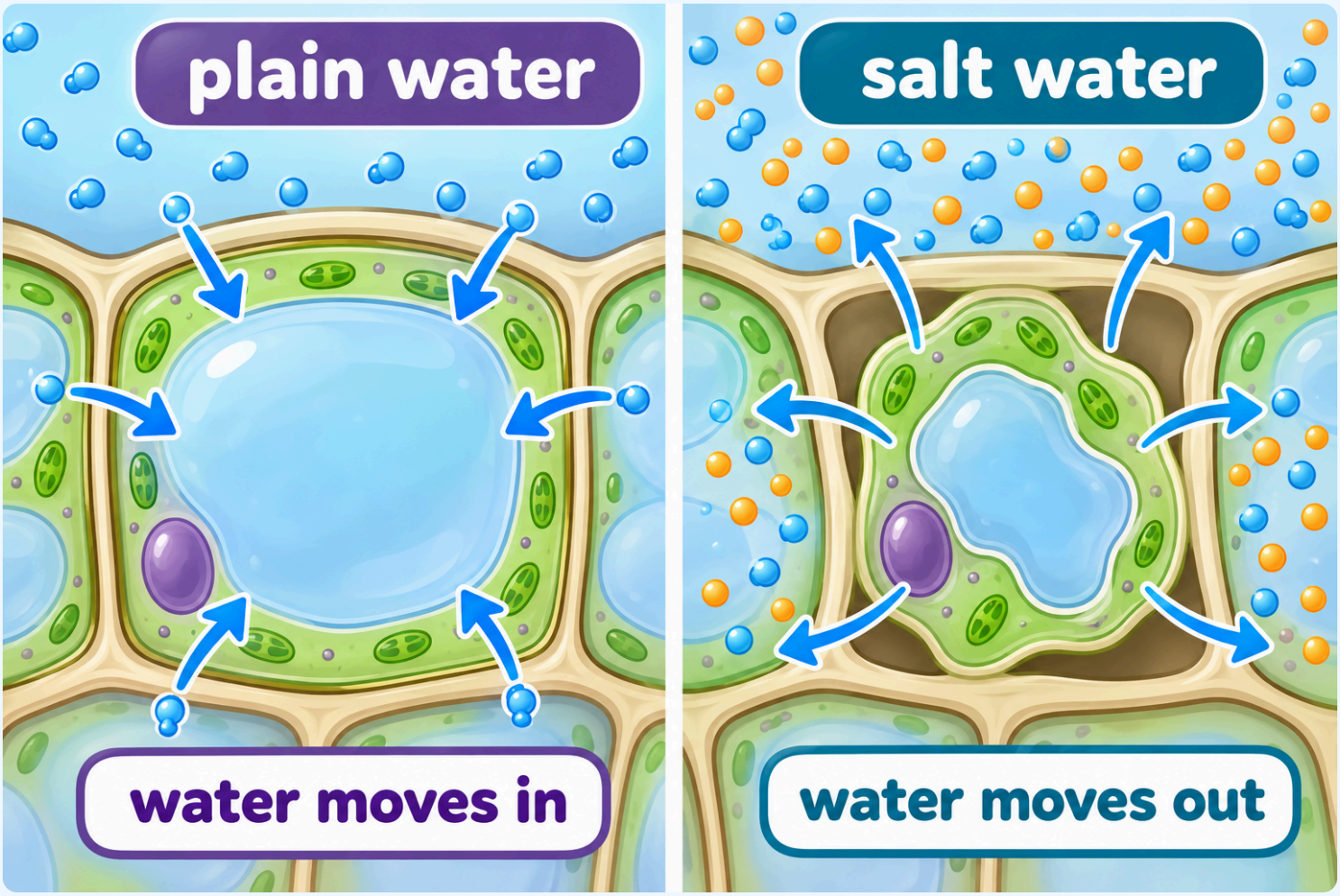


An independent investigation: plan carefully, collect evidence, and let your results do some arguing.

Potato prep: skin OFF; equal strips only



Ask an adult to peel the potato, then cut strips from the peeled interior: about 1 cm × 1 cm × 5 cm. Use the same size for every cup.



A model for the invisible part: water movement changes how firm the potato cells become.

📋 **NGSS Standard:**
MS-LS1-2 — Develop and use a model to describe the function of a cell as a whole way and ways parts of cells contribute to the function.

🎯 Learning Objective

Students will plan and conduct a fair investigation, organize evidence, and use scientific reasoning to explain a pattern in their results.

💡 The Big Idea

Plant-cell membranes let some substances cross more easily than others. Water can move across the membrane by osmosis, changing the firmness and mass of plant tissue.

🔬 The Science

Selective membrane

A cell membrane is not an open door. It controls movement into and out of the cell, helping keep conditions workable.

Osmosis

Water moves by osmosis toward the side with more dissolved particles. In a plant cell, water moving in can make the cell more turgid; water moving out can make it less firm.

A useful model

A potato strip contains many plant cells. It is not one giant cell, but changes in its mass or firmness give evidence about water movement across many cell membranes.

💡 **Key Vocabulary**

cell membrane · **selectively permeable** · **osmosis** · **solute** · **solution** · **concentration** · **turgid** · **mass**

What you are actually testing

Independent variable

The one factor you deliberately change: **salt-solution concentration**.

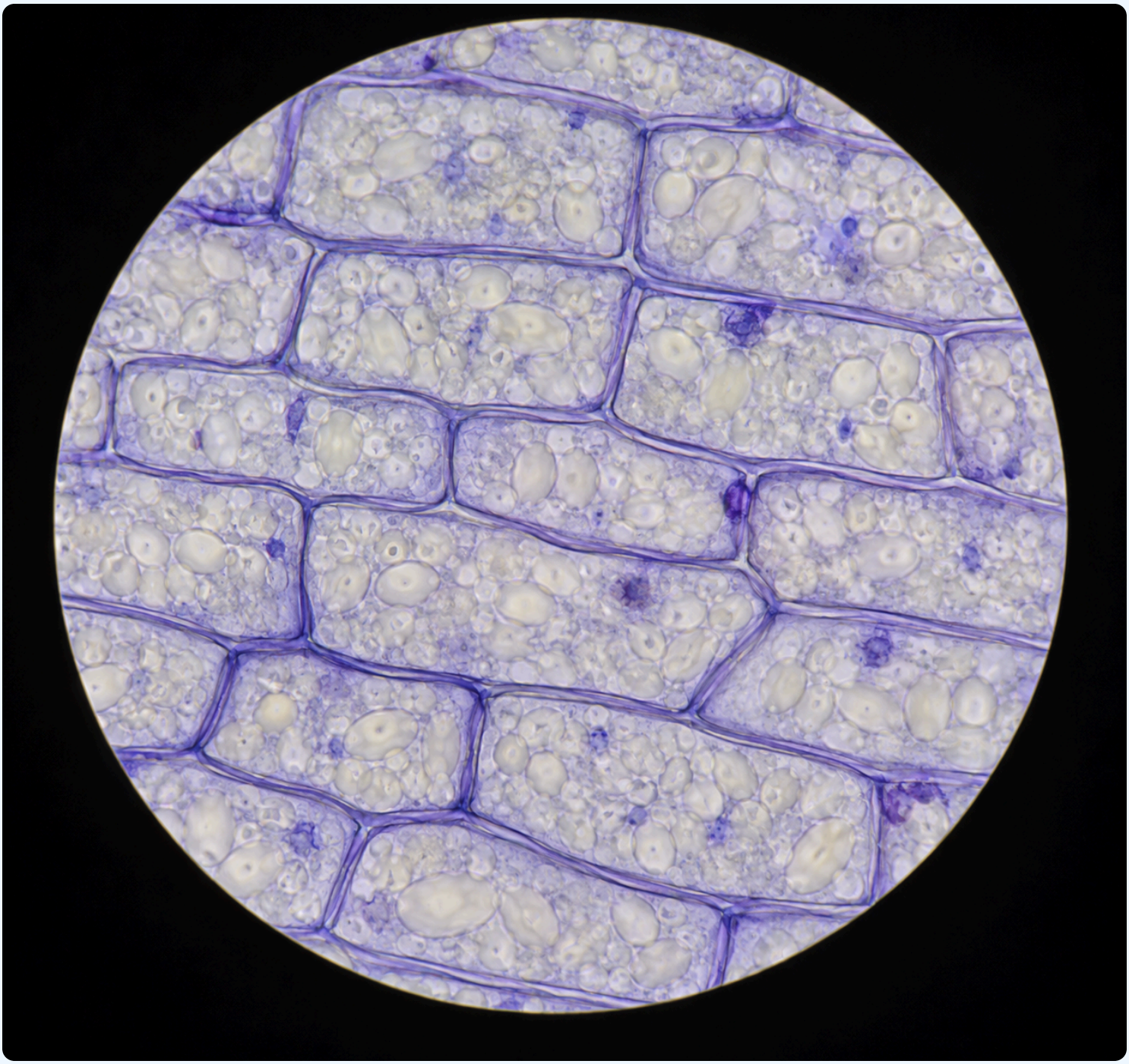
Dependent variable

What you measure because it may respond: **change in potato mass or length, plus firmness**.

Controls

Conditions you keep the same so the comparison is fair: **strip size, liquid volume, soak time, and temperature**.

Potato cells under a microscope slide



Illustrative microscope view: the rectangular outlines are cell walls. Each strip contains many cells, not one giant cell.

📦 Materials

🏠 Household Items to Gather

- 1 medium potato — **skin removed**; an adult cuts equal peeled strips, about **1 cm × 1 cm × 5 cm**
- 3–4 clear cups
- water and table salt
- measuring spoons and labels
- kitchen scale or ruler
- paper towel

📋 Printables

- This lesson worksheet
- Printable reference card
- Science notebook or loose paper

🧪 Let’s Get Started

Your investigation question
How does salt-solution concentration affect the mass or firmness of potato tissue?
Independent variable: salt-solution concentration
Dependent variable: change in mass (g) or length (mm), plus firmness
Controls: potato-strip size, solution volume, soak time, temperature

Step 1 — Prepare the potato correctly.
Ask an adult to peel the potato completely, then cut equal strips from the peeled interior: about **1 cm × 1 cm × 5 cm**. Keep the skin off every strip. Label cups with a control (plain water) and your chosen salt concentrations.

Step 2 — Measure and record each strip’s starting mass or length.
Make a prediction before the strips enter the cups.

Step 3 — Add the same volume of solution to every cup.
Place one strip in each cup and record the start time.

Step 4 — After at least 24 hours, blot each strip gently in the same way.
Measure again and record firmness observations.

Step 5 — Calculate change: after − before.
If using mass, calculate percent change and graph concentration against change.

💬 Discussion Questions

Which direction did water move in your plain-water condition? *Often into the potato cells, if the strip gained mass or became firmer. Ask the student to connect their evidence to the claim.*

Why must strips start the same size? *Different-sized strips have different amounts of tissue and surface area, making the comparison unfair.*

What does “more concentrated” mean? *It means more dissolved solute particles in a given volume of solution.*

Why record mass as well as firmness when possible? *Mass gives a numerical measurement; firmness is useful qualitative evidence. Together they make a stronger case.*

What result would make you revise your explanation? *A trend that reverses, no pattern after careful trials, or conflicting measurements should prompt another test.*

🚀 Extensions

Stretch your thinking: Calculate percent change in mass: (after − before) ÷ before × 100. Predict the approximate isotonic concentration, where there is little net water movement. Explain why percent change makes comparisons fairer when strips do not begin with exactly the same mass.

👩‍🏫 Parent / Teacher Notes

Keep the independence real: Give the student the question, safety boundaries, and materials; let them make the prediction, decide a reasonable measurement, and defend their controls. Ask “What evidence makes you say that?” before supplying an answer.

Useful video: Potato Osmosis Experiment (Science Class) shows a potato-piece comparison in water and salt solution. Watch it after making your prediction so the student’s first explanation stays their own.

📝 Student Investigation Worksheet

Name: _____ **Date:** _____

Question, variables, and prediction

Question: How does salt-solution concentration affect the mass or firmness of potato tissue?

Independent variable: _____ **Dependent variable:** _____

Controls: _____

Prediction: If _____, then _____ because _____.

Evidence Table

Solution	Before mass / length	After mass / length	Change	Firmness evidence

Graph Your Pattern

Give your graph a title. Label both axes and include units.

Claim–Evidence–Reasoning

Claim: What pattern did you find?

Evidence: Use at least two measurements or observations.

Reasoning: Use the science vocabulary to explain why the evidence supports your claim. Name one uncertainty or improvement.

✂️ PARENT ANSWER KEY — Detach before giving worksheet to students

Expected results

In plain water, strips often gain mass or feel firmer. In more concentrated salt solutions, they often lose mass or feel softer. Real potatoes vary, so the student's evidence and controls matter more than an exact number.

What mastery looks like

- Identifies a testable variable and protects meaningful controls.
- Records repeated observations or measurements accurately.
- Uses a graph or table to state a claim.
- Connects evidence to the scientific principle instead of only describing what happened.

Troubleshooting

If results are inconsistent, check whether the student changed more than one variable, used unequal starting materials, or measured at different times. A surprising result is not a failed experiment; repeat it and improve the test.

Printable reference card → · Collection home